C Piscine

C 08

Summary: This document is the subject for C 08 module of the C Piscine @ 42.

Version: 12.2

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Chapter I

Instructions

- Only this page will serve as reference: do not trust rumors.
- Watch out! This document could potentially change before submission.
- Make sure you have the appropriate permissions on your files and directories.
- You have to follow the submission procedures for all your exercises.
- Your exercises will be checked and graded by your fellow classmates.
- On top of that, your exercises will be checked and graded by a program called Moulinette.
- Moulinette is very meticulous and strict in its evaluation of your work. It is entirely automated and there is no way to negotiate with it. So if you want to avoid bad surprises, be as thorough as possible.
- Moulinette is not very open-minded. It won't try and understand your code if it doesn't respect the Norm. Moulinette relies on a program called `norminette` to check if your files respect the norm. TL;DR: it would be idiotic to submit a piece of work that doesn't pass `norminette`'s check.
- These exercises are carefully laid out by order of difficulty - from easiest to hardest. We `will not` take into account a successfully completed harder exercise if an easier one is not perfectly functional.
- Using a forbidden function is considered cheating. Cheaters get `-42`, and this grade is non-negotiable.
- You'll only have to submit a main() function if we ask for a program.
- Moulinette compiles with these flags: -Wall -Wextra -Werror, and uses `cc`.
- If your program doesn't compile, you'll get `0`.
- You cannot leave any additional file in your directory than those specified in the subject.
- Got a question? Ask your peer on the right. Otherwise, try your peer on the left.

- Your reference guide is called `Google / man / the Internet / ....`
- Check out the "C Piscine" part of the forum on the intranet, or the slack Piscine.
- Examine the examples thoroughly. They could very well call for details that are not explicitly mentioned in the subject...
- By Odin, by Thor ! Use your brain !!!

Chapter II

Foreword

Here's what `Wikipedia` have to say about `Platypus`:

```
The platypus (Ornithorhynchus anatinus), also known as the duck-billed platypus,
is a semiaquatic egg-laying mammal endemic to eastern Australia,
including Tasmania. Together with the four species of echidna, it is one of
the five extant species of monotremes, the only mammals that lay eggs instead
of giving birth. The animal is the sole living representative of its family
(Ornithorhynchidae) and genus (Ornithorhynchus), though a number of related
species have been found in the fossil record.

The unusual appearance of this egg-laying, duck-billed, beaver-tailed,
otter-footed mammal baffled European naturalists when they first encountered
it, with some considering it an elaborate hoax. It is one of the
few venomous mammals, the male platypus having a spur on the hind foot that
delivers a venom capable of causing severe pain to humans. The unique features
of the platypus make it an important subject in the study of evolutionary
biology and a recognisable and iconic symbol of Australia; it has appeared
as a mascot at national events and is featured on the reverse of its 20-cent
coin. The platypus is the animal emblem of the state of New South Wales.

Until the early 20th century, it was hunted for its fur, but it is now
protected throughout its range. Although captive breeding programs have
had only limited success and the platypus is vulnerable to the effects
of pollution, it is not under any immediate threat.
```

This subject is absolutly not talking about platypus.

Chapter III

Exercise 00 : ft.h

	Exercise 00
ft.h	
Turn-in directory : $ex00/$	
Files to turn in : `ft.h`	
Allowed functions : `None`	

- Create your `ft.h` file.
- It should countain the prototypes of all the following functions:

```
void    ft_putchar(char c);
void    ft_swap(int *a, int *b);
void    ft_putstr(char *str);
int     ft_strlen(char *str);
int     ft_strcmp(char *s1, char *s2);
```

Chapter IV

Exercise 01 : ft_boolean.h

	Exercise 01
	ft_boolean.h
Turn-in directory : *ex*01/	
Files to turn in : `ft_boolean.h`	
Allowed functions : `None`	

- Create a `ft_boolean.h` file. It'll compile and run the following main appropriately:

```
#include "ft_boolean.h"

void        ft_putstr(char *str)
{
        while (*str)
                write(1, str++, 1);
}

t_bool        ft_is_even(int nbr)
{
        return ((EVEN(nbr)) ? TRUE : FALSE);
}

int                main(int argc, char **argv)
{
        (void)argv;
        if (ft_is_even(argc - 1) == TRUE)
                ft_putstr(EVEN_MSG);
        else
                ft_putstr(ODD_MSG);
        return (SUCCESS);
}
```

- This program should display

```
I have an even number of arguments.
```

- or

```
I have an odd number of arguments.
```

- followed by a line break when adequate.

`Norminette must be launched with the -R CheckDefine flag.`
`Moulinette will use it too.`

Chapter V

Exercise 02 : ft_abs.h

	Exercise 02
ft_abs.h	
Turn-in directory : $ex02/$	
Files to turn in : `ft_abs.h`	
Allowed functions : `None`	

- Create a macro `ABS` which replaces its argument by it absolute value:

```
#define ABS(Value)
```

> Norminette must be launched with the *-R CheckDefine* flag.
> Moulinette will use it too.

Chapter VI

Exercise 03 : ft_point.h

	Exercise 03
ft_point.h	
Turn-in directory : $ex03/$	
Files to turn in : `ft_point.h`	
Allowed functions : `None`	

- Create a file `ft_point.h` that'll compile the following main:

```
#include "ft_point.h"

void        set_point(t_point *point)
{
        point->x = 42;
        point->y = 21;
}

int                main(void)
{
        t_point                point;

        set_point(&point);
        return (0);
}
```

Chapter VII

Exercise 04 : ft_strs_to_tab

	Exercise 04
ft_strs_to_tab	
Turn-in directory : $ex04/$	
Files to turn in : `ft_strs_to_tab.c`	
Allowed functions : `malloc, free`	

- Create a function that takes an array of string as argument and the size of this array.
- Here's how it should be prototyped:

```
struct s_stock_str *ft_strs_to_tab(int ac, char **av);
```

- It will transform each element of av into a structure.
- The structure will be defined in the `ft_stock_str.h` file that we will provided, like this:

```
typedef struct s_stock_str
{
  int  size;
  char *str;
  char *copy;
}            t_stock_str;
```

  - `size` being the length of the string;
  - `str` being the string;
  - `copy` being a copy of the string;
- It should keep the order of av.

- The returned array should allocated in memory and its last element's `str` set to 0, this will mark the end of the array.
- It should return a NULL pointer if an error occurs.
- We'll test your function with our `ft_show_tab` (next exercise). Make it work according to this !

Chapter VIII

Exercise 05 : ft_show_tab

	Exercise 05
ft_show_tab	
Turn-in directory : $ex05/$	
Files to turn in : `ft_show_tab.c`	
Allowed functions : `write`	

- Create a function that displays the content of the array created by the previous function.
- Here's how it should be prototyped:

```
void ft_show_tab(struct s_stock_str *par);
```

- The structure will be the same as the previous exercise and will be defined in the `ft_stock_str.h` file
- For each element, we'll display:
  - the string followed by a '\n'
  - the size followed by a '\n'
  - the copy of the string (that could have been modified) followed by a '\n'
- We'll test your function with our `ft_strs_to_tab` (previous exercise). Make it work according to this !

Chapter IX

Submission and peer-evaluation

Turn in your assignment in your `Git` repository as usual. Only the work inside your repository will be evaluated during the defense. Don't hesitate to double check the names of your files to ensure they are correct.

> ⚠ `You need to return only the files requested by the subject of this project.`